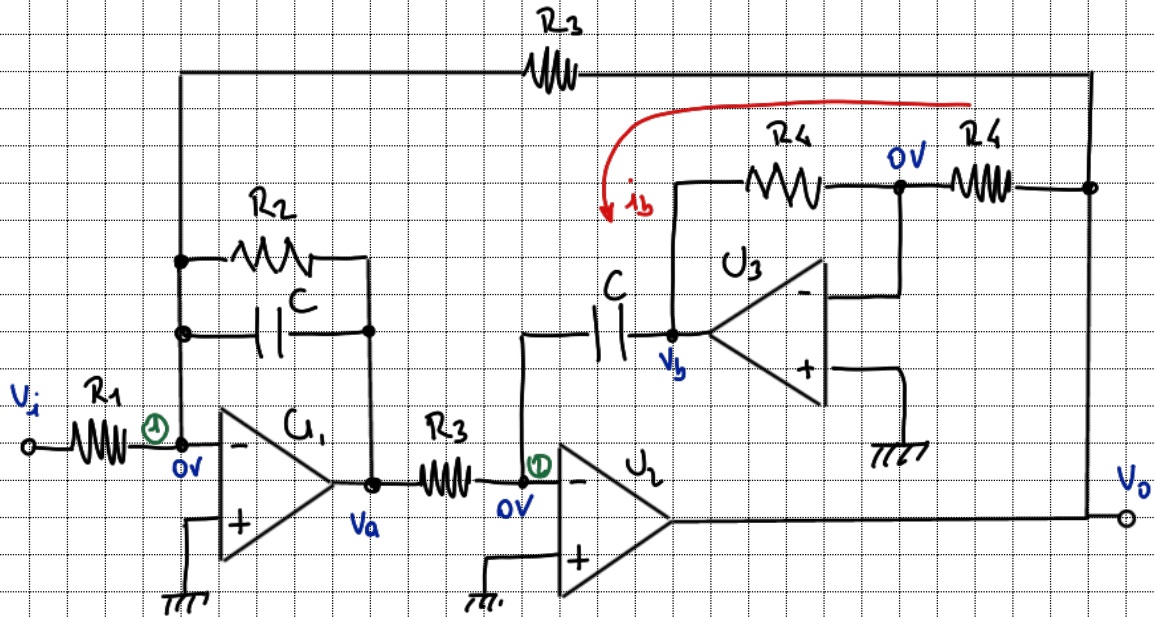


Tarea 2 - Montenegro, Maximiliano



$$\textcircled{1} V_i G_1 = -V_a (SC + G_2) - V_o G_3$$

$$\textcircled{1} 0 = V_a G_3 + V_b SC$$

$$-V_a G_3 = V_b SC$$

$$-V_a G_3 = -V_o SC$$

$$V_a G_3 = V_o SC$$

$$\textcircled{2} V_a = V_o S R_3 C$$

$$\textcircled{2} \rightarrow \textcircled{1}$$

$$V_i G_1 = -V_o S R_3 C (SC + G_2) - V_o G_3$$

$$V_i G_1 = -V_o (S R_3 C (SC + G_2) - G_3)$$

$$T(s) = \frac{V_o}{V_i} = - \frac{G_1}{S R_3 C (SC + G_2) - G_3} = - \frac{1}{R_1} \frac{1}{S^2 R_3 C^2 + SC R_3 G_2 - G_3}$$

$$T(s) = - \frac{1}{R_1} \frac{1}{S^2 R_3 C^2 + SC \frac{1}{R_2} - \frac{1}{R_3}} = - \frac{R_2 R_3}{R_1} \frac{1}{S^2 C^2 R_3^2 R_2 + SC R_3^2 - R_2}$$

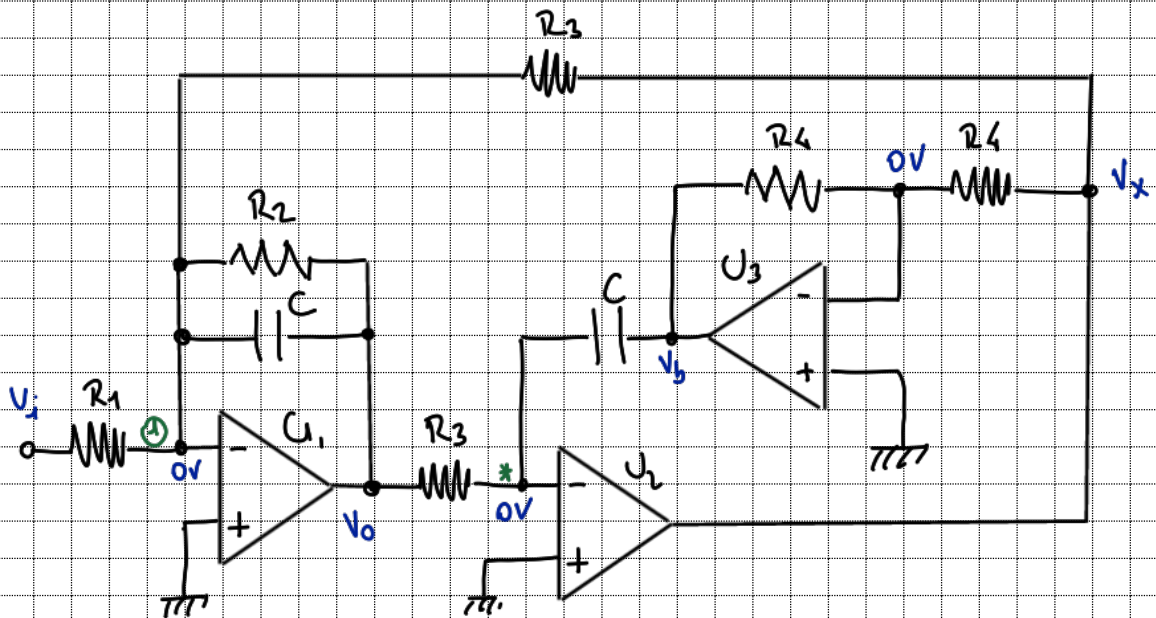
$$T(s) = - \frac{R_2 R_3}{R_1} \cdot \frac{1}{C^2 R_3^2 R_2} \frac{1}{S^2 + S \frac{1}{R_2 C} + \frac{1}{R_3^2 C^2}}$$

$$T(s) = - \frac{1}{R_1 R_3 C^2} \frac{1}{S^2 + S \frac{1}{R_2 C} + \frac{1}{(R_3 C)^2}}$$

$$\omega_0 = \frac{1}{R_3 C}$$

$$K = - \frac{R_3}{R_1}$$

$$\frac{\omega_0}{Q} = \frac{1}{R_2 C} \Rightarrow Q = R_2 C \omega_0 \Rightarrow Q = \frac{R_2 C}{R_3 C} \Rightarrow Q = \frac{R_2}{R_3}$$



$$\begin{cases} V_i G_1 = -V_0 (G_2 + SC) - V_x G_3 & * \quad V_0 G_3 = -V_b SC \\ V_0 G_3 = V_x SC & -V_x = V_b \end{cases}$$

$$V_0 \frac{G_3}{SC} = V_x$$

$$V_i G_1 = -V_0 (G_2 + SC) - V_0 \frac{G_3}{SC} \cdot G_3$$

$$V_i G_1 = -V_0 \left(\frac{SC(G_2 + SC) + G_3^2}{SC} \right) = \frac{V_0}{V_i} = \frac{-G_1 SC}{SC(G_2 + SC) + G_3^2}$$

$$\frac{V_0}{V_i} = T(s) = -\frac{1}{R_1 C} \frac{SC}{(SC)^2 + SCG_2 + G_3^2} = \frac{1}{R_1 C} \frac{SC}{s^2 + s \frac{1}{R_2 C} + \frac{1}{(R_3 C)^2}}$$

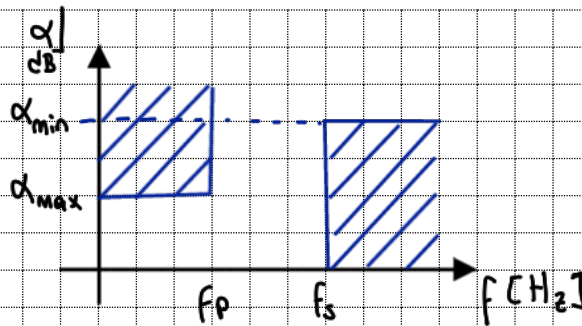
$$T(s) = -\frac{1}{R_1 C} \frac{s}{s^2 + s \frac{1}{R_2 C} + \frac{1}{(R_3 C)^2}}$$

$$\omega_0^2 = \frac{1}{(R_3 C)^2} \Rightarrow \omega_0 = \frac{1}{R_3 C} \quad K = \frac{1}{R_1 C}$$

$$\frac{\omega_0}{Q} = \frac{1}{R_2 C} \Rightarrow Q = \omega_0 R_2 C \Rightarrow Q = \frac{1}{R_3 C} R_2 C \Rightarrow Q = \frac{R_2}{R_3}$$

los parámetros son los siguientes

$$\omega_0 = \frac{1}{R_3 C} ; Q = \frac{R_2}{R_3} ; K = \frac{1}{R_1 C}$$



$$\alpha_{\max} = 0,4 \text{ dB}$$

$$\alpha_{\min} = 48 \text{ dB}$$

$$f_p = 3200 \text{ Hz}$$

$$f_s = 9600 \text{ Hz}$$

$$f_p' = \frac{f_p}{f_p} = 1 ; f_s' = \frac{f_s}{f_p} = 3$$

$$(f_p' = 1 \quad f_s' = 3) 2\pi \rightarrow \Omega [\text{rad/s}]$$

$$\alpha = \sqrt{1 + \epsilon^2 \cosh^2(n \cosh^{-1}(w))}$$

$$\alpha \Big|_{\text{dB}} = 20 \log(1 + \epsilon^2 \cosh^2(n \cosh^{-1}(w)))$$

$$\alpha \Big|_{\text{dB}} = 10 \log(1 + \epsilon^2 \cosh^2(n \cosh^{-1}(w)))$$

$$\frac{\alpha \Big|_{\text{dB}}}{10} - 1 = \epsilon^2 \cosh^2(n \cosh^{-1}(w))$$

con $w=1$

$$\cosh^{-1}(1) = 0 \Rightarrow \cosh^2(0) = 1$$

$$\epsilon^2 = \frac{\alpha_{\max} \Big|_{\text{dB}}}{10} - 1$$

$$\epsilon^2 = 10^{0,4/10} - 1$$

$$\epsilon^2 = 1,09648 - 1$$

$$\epsilon^2 = 0,09648$$

$$\epsilon = 0,31061$$

con $w=3$

n	α
1	2,71
2	4,21
3	39,91
4	45,068
5	70,53454

$$C_0 = 1$$

$$C_1 = w$$

$$C_n = 2wC_{n-1}(w) - C_{n-2}(w)$$

$$C_2 = 2w(w) - 1 = 2w^2 - 1$$

$$C_3 = 2w(2w^2 - 1) - w = 4w^3 - 2w - w = 4w^3 - 3w$$

$$C_4 = 2w(4w^3 - 3w) - (2w^2 - 1) = 8w^4 - 6w^2 - 2w^2 + 1 = 8w^4 - 8w^2 + 1$$

$$C_5 = 2w(8w^4 - 8w^2 + 1) - (4w^3 - 3w) = 16w^5 - 16w^3 + 2w - 4w^3 + 3w$$

$$C_5 = 16w^5 - 20w^3 + 5w$$

$$\begin{aligned}
 T(s) \cdot T(-s) &= \frac{1}{1 + \epsilon^2 (C_s(\omega))^2} \\
 &= \frac{1}{1 + \epsilon^2 (16\omega^5 - 20\omega^3 + 5\omega)^2} \rightarrow (16\omega^5 - (20\omega^3 - 5\omega))^2 \\
 &= \frac{1}{1 + \epsilon^2 (256\omega^{10} - 2(16\omega^5)(-20\omega^3 - 5\omega) + (20\omega^3 - 5\omega)^2)} \\
 &= \frac{1}{1 + \epsilon^2 (256\omega^{10} - 640\omega^8 + 160\omega^6 + 400\omega^6 - 200\omega^4 + 25\omega^2)} \\
 &= \frac{1}{1 + \epsilon^2 (256\omega^{10} - 640\omega^8 + 560\omega^6 - 200\omega^4 + 25\omega^2)}
 \end{aligned}$$

$$s = j\omega \rightarrow \omega = \frac{s}{j} \rightarrow \omega = -js \rightarrow \omega^2 = (-js)^2 \rightarrow \omega^2 = (-j)^2 (j)^2 s^2 \rightarrow \omega^2 = -s^2$$

$$\Rightarrow \omega^2 = -s^2; \quad \omega^4 = s^4; \quad \omega^6 = -s^6; \quad \omega^8 = s^8; \quad \omega^{10} = -s^{10}$$

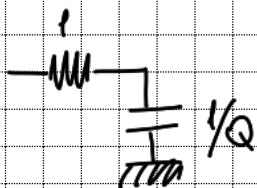
$$= \frac{1}{1 + \epsilon^2 (-s^{10} 256 - s^8 640 - s^6 560 - s^4 200 - s^2 25)}$$

$$T(s) \cdot T(-s) = \frac{1}{\epsilon^2 256} \frac{1}{s^{10} + s^8 \frac{640}{256} + s^6 \frac{560}{256} + s^4 \frac{200}{256} + s^2 \frac{25}{256} - \frac{1}{\epsilon^2 256}}$$

Mediante el script

$$T(s) = \frac{1}{s^5 + 3,236 s^4 + 5,236 s^3 + 5,236 s^2 + 3,236 s + 1} \cdot \frac{1}{16\epsilon}$$

$$T(s) = \frac{1}{s+1} \cdot \frac{1}{s^2 + s \cdot 2 \cos\left(\frac{\pi}{5}\right) + 1} \cdot \frac{1}{s^2 + s \cdot 2 \cos\left(\frac{3\pi}{5}\right) + 1}$$



$$Q = \frac{1}{2 \cos \varphi}$$

$$Q_1 = \frac{1}{2 \cos \frac{\pi}{5}} = 0,618 = Q_1$$

$$Q_2 = \frac{1}{2 \cos \frac{3\pi}{5}} = 1,618 = Q_2$$

$$C = 10 \text{ nF}$$

$$\frac{1}{s+1} \Rightarrow \omega = 1 \Rightarrow R = \frac{1}{2\pi f_c C} = 4973,6$$

$$K = \frac{1}{16\epsilon} = \frac{1}{16 \cdot 0,31061} = 0,2012 = K$$

$$\Rightarrow T(s) = \frac{1}{s+1} \cdot \frac{1}{s^2 + s \cdot \frac{1}{0,618} + 1} \cdot \frac{1}{s^2 + s \cdot \frac{1}{1,618} + 1}$$

$$T(s) = - \frac{1}{R_1 R_3 C^2} \frac{1}{s^2 + s \cdot \frac{1}{R_2 C} + \frac{1}{(R_3 C)^2}}$$

$$f = s\omega_0$$

$$T(s) = - \frac{1}{R_1 R_3 C^2} \frac{1}{(s\omega_0)^2 + s\omega_0 \frac{1}{R_2 C} + \frac{1}{(R_3 C)^2}}, \quad \frac{1}{R_3 C} = \omega_0$$

$$= - \frac{R_3 \omega_0^2}{R_1} \frac{1}{s^2 \omega_0^2 + s \omega_0^2 \frac{R_2}{R_1} + \omega_0^2}$$

$$= - \frac{R_3}{R_1} \frac{\cancel{\omega_0^2}}{\cancel{\omega_0^2} s^2 + s \frac{1}{\frac{R_1}{R_2}} + 1} \quad \frac{R_1}{R_2} = Q$$

$$= - \frac{R_3}{R_1} \frac{1}{s^2 + s \frac{1}{Q} + 1} \quad ; \quad K = - \frac{R_3}{R_1} \quad ; \quad Q = \frac{R_1}{R_2} \quad ; \quad \omega_0^2 = \frac{1}{(R_3 C)^2}$$

elijo $C = 10 \text{ nF}$

$$\Rightarrow \omega_0 = \frac{1}{R_3 C} \quad \therefore R_3 = \frac{1}{\omega_0 C} = \frac{1}{2\pi \cdot 3200 \cdot 10 \cdot 10^{-9}} \Rightarrow R_3 = 4973,6$$

$$Q = \frac{R_1}{R_2} \Rightarrow R_1 = Q \cdot R_2 \Rightarrow R_2 = 3073,7 \Omega$$

$$|K| = \frac{R_3}{R_1} \Rightarrow R_1 = \frac{R_3}{|K|} = \frac{4973,6}{0,2012} = 24719,68 \Omega = R_1$$

$$Q_2 = \frac{R_{21}}{R_3} \Rightarrow R_{21} = Q_2 R_3 = 8047,27 \Omega = R_{21}$$

$$R_1 \quad 24719,68 \Omega$$

$$C = 10 \text{ nF}$$

$$R_2 \quad 3073,7 \Omega \quad | \quad 8047,27 \Omega$$

$$R_3 \quad 4973,6 \Omega$$

$$R = 4973,6 \Omega$$

$$R_4 \quad 1k$$